

Total No. of Questions : 6]

SEAT No. :

P5114

[Total No. of Pages : 3

BE/Insem - 521
B.E. (Mechanical) (Semester - I)
REFRIGERATION & AIR CONDITIONING
(2012 Pattern)

Time :1 hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Figures to the right side indicate full marks.

Q1) a) List the various processes in Bell Coleman cycle. Draw cycle on P-v and T-s diagram. Derive expression of COP of Bell Coleman Cycle. [5]

b) Explain the need of air conditioning for multiplex facilities. [5]

OR

Q2) a) With neat schematic, explain the refrigeration system for Ice plant. [5]

b) Dense air refrigeration system of 10 TR capacity operates between pressure of 4 bar and 16 bar. The air temperature after heat rejection to surrounding is 37°C and air temperature at exit of refrigerator is 7 °C. Determine (i) Compressor and turbine work per TR; (ii) C.O.P. (iii) Power required per TR. Take $n = 1.25$, and $C_p = 1 \text{ kJ/kg K}$. [5]

Q3) a) Explain the benefits of secondary refrigerants. What is selection criteria of secondary refrigerants? [4]

b) A simple R - 134a heat pump for space heating operates between temperature limits of 15 °C and 50 °C. The heat required to pumped is 100 MJ/h. Determine. [6]

i) Quality at evaporator inlet. ii) the discharge temperature (assume C_p of vapour 0.996 kJ/kg.K), iii) piston displacement of compressor,

iv) Compressor power input & v) Cop of heating. The specific volume of R-134a saturated vapour at 15 °C is 0.0418 m³/ kg.

P.T.O.

Saturation Temperature (°C)	Pressure, MPa	Enthalpy, kJ/kg		Entropy, kJ/kg K	
		hf	hg	sf	sg
15	0.488	220.4	407.1	1.0725	1.72
50	1.318	271.6	423.4	1.24	1.707

OR

Q4) A refrigerant R - 134a vapour compression includes a liquid to vapour heat exchanger in the system. The heat exchanger cools saturated liquid coming out of condenser from 32 °C to 22 °C with the help of vapour coming out of evaporator at -12 °C saturated. The compression is isentropic. Draw the line diagram of components, represents the processes on p -h diagram and find:
 (i) C.O.P. of the system; (ii) Refrigerating capacity of system if the compressor displacement is 1.2m³/min (iii) C.O.P. of the system without the heat exchanger. Take C_{pl} at 32 °C (Sat) = 1.456 KJ/kg K & C_{pg} at - 12 °C (sat) = 0.8463 kJ/kg K

Q5) A Multi - evaporator, vapour compression refrigeration system is working with R - 134a refrigerant. The system uses individual expansion valve arrangement. The refrigeration capacity of the high temperature evaporator operating at - 10 °C is 5 TR, while it is 10 TR for the low temperature evaporator operating at - 35 °C. The condenser pressure is 11bar. Separate compressors are used with each evaporators. Assume saturated conditions at the exit of both evaporators and condenser, and isentropic compression. Draw schematic of system. [10]

- i) Find the required power input to compressor in kW
- ii) COP.

OR

- Q6)** a) With neat diagram explain Cascade refrigeration system. [5]
 b) Draw the neat schematic of practical Li-Br absorption system. List the various components with their function in brief. [5]



